LayerForge v9 — Appendix

Dataset Catalog, Verification Records, Parameter Ablation, and Pareto Plots

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Appendix A. Dataset Catalog

All datasets used in v9 verifications, with language, domain, N range, source, and licensing notes. Datasets are read-only mounted at C:/work/dataset-vault/readonly/ for verification reproducibility.

Dataset	Language	Domain	Length	N used	Source
WildChat (allenai/WildChat)	EN	multi-turn dialogue	mid (1-5k)	3-100	HuggingFace
ShareGPT	EN	multi-turn dialogue	mid	3-10	ShareGPT_V3_unfiltered
LongBench hotpotqa	EN	multi-hop QA	long (50-100k)	5-100	LongBench v1
loogle	EN	long single-doc	long (10k+)	5	loogle
CNN/DailyMail v3	EN	news summary	mid	10	HuggingFace cnn_dailymail
livedoor news	JA	news 9 categories	mid (1-2k)	9-27	livedoor-news-corpus
jmultiwoz	JA	dialogue	short	10	jmultiwoz
jawiki	JA	wiki	long (10k+)	5 synthetic	jawiki_singletongue
Aozora	JA	literature	long (10k+)	5	aozorabunko_clean
BSD parallel	EN/JA	dialogue parallel	short	5 each path	bsd_ja_en
meetingbank	EN	meeting transcript	mid-long	5	meetingbank
mt_bench_101	EN	dialogue	short	10	mt_bench_101
polbooks (Newman)	graph	community K=3	105 nodes	1	newman_adjnoun benchmark
football (Newman)	graph	community K=12	115 nodes	1	newman benchmark

Table A1. Dataset catalog (14 entries).

A.1 Sampling Protocols

Sampling protocols vary by verification: deterministic head (V-018, V-021-042 hotpotqa), stratified by category (V-013-014 livedoor 9 cat), random with seed (V-005 claim1 robustness), synthetic grouped (V-017 jawiki 200-sentence units due to sentence-level original structure). Sampling protocol is recorded per V-ID in Appendix B.

Appendix B. Verification Record Summary

Complete verification record list (46 entries: Table B1 V-001-V-020 = 20 rows, Table B2 V-021-V-042-quad + 1 cross-corpus = 26 rows). Each row shows V-ID, dataset, N, verdict, and commit hash for traceability. Full Sections 1-8 records are in docs/verifications/*.md.

V-ID	Dataset	N	Verdict (key metric)	Commit
V-001	WildChat (Phase 2b entry)	962/50K	PASS dataset feasibility	(retro)
V-002	WildChat	10	PASS pipeline shape (F1 cosine 0.538)	(retro)
V-003	WildChat (step_a)	10	format artifact confirmed (F4 0.823 > F1 0.538)	(retro)
V-004	WildChat (claim1)	100	formal PASS reduction 93.00% mean	76a8bda
V-005	WildChat (robustness)	100	3-axis PASS (random/survivorship/length)	a13ce3f
V-006	WildChat (claim2 F3)	3	FAIL (refusal 33.3%, specific divergent)	36ced95
V-007	WildChat (claim2 F4 hybrid)	6	strong PASS (refusal 0/6, cosine 0.939)	f95010f
V-008	polbooks	105 nodes	AMBIGUOUS (ARI 0.6140, K=6 over-seg)	d88804f
V-009	football	115 nodes	K-FAIL but ARI 0.8549 high	991671e
V-010	mt_bench_101 + jmultiwoz	10 each	MIXED (mt PASS 77% / jmulti FAIL 16%)	3e68346
V-011	loogle	5	scope-FIT extreme 99% + caveat	13d01a9
V-012	meetingbank	5	intermediate + length sensitivity	af98808
V-013	livedoor (JA default)	9 stratified	near-zero 2.5% chars (JA + default mismatch)	bb976c7
V-014	livedoor (JA + MeCab)	9 stratified	improvement-LARGE delta +83.89 pp (86.40%)	fb51faf
V-015	jmultiwoz + MeCab	10	improvement-LARGE delta +59.46 pp (75.48%)	fb182fb
V-016	14 dataset length regression	8 dataset	v7 partial confirm (R-sq=0.54, p=0.023)	79b5cce
V-017	jawiki	5 synthetic	v7-fit INTERMEDIATE 93.65% + sigmoid hint	ffb5597
V-018	ShareGPT + LongBench	3+5	WildChat-confirm + loogle-confirm	b341cc5
V-019	Aozora + BSD parallel	5 + 5x3	ceiling 99.29% + MeCab>EN asymmetry	547f7aa
V-020	ShareGPT + hotpotqa	3+5	LayerForge-COMPETITIVE vs LDA/NMF	1dee6cd

Table B1. Verification records V-001 through V-020 (Phase 2a + 2b base).

V-ID	Dataset	N	Verdict (key metric)	Commit
V-021	hotpotqa (factual recall)	10	PRESERVE-FAIL substring 10%, tok 18%	09d44bc
V-022	livedoor (ROUGE-L)	9	RETENTION-FAIL R-L 0.074, delta +0.113	5e0f21a
V-023	hotpotqa attribute breakdown	10 retro	ATTR-TYPE-UNIFORM-FAIL + CONTEXT-DOMINANT	e51a847
V-024	hotpotqa F3 downstream	5	REFUSAL-CONSTITUTIVE 5/5 (V-006 N=5 stronger)	c5a53d4
V-024-bis	hotpotqa F4 hybrid downstream	5	REFUSAL-AVOIDED + ACCURACY-WEAK 0/5	ea86ac4
V-025	WildChat retro BERTScore	6	roberta 0.927 HIGH / mbert 0.838 MID-HIGH	d3c678b
V-026	ShareGPT+hotpotqa topic coherence	6	LF-COMPETITIVE NPMI + UMass best 5/6 (hotpot/4 NMF outlier)	79bff78
V-027	hotpotqa tokenizer ext ablation	10	NEGATIVE (driver-level fix fails)	bbe1b89

V-ID	Dataset	N	Verdict (key metric)	Commit
V-029-a	hotpotqa cost+latency	5	Cost PASS 99%/84%, Latency FAIL -3.6%	59bb20a
V-029-b	livedoor cross-doc overlap	27	DISCRIMINATIVE-STRONG ratio 1.31x	783327b
V-029-c	ShareGPT theme baton-pass	10	BATON-PASS-STRONG ratio 1.99x	7cebd7b
V-029-d	livedoor title BERTScore	27	TRIAGE-PROXY-RELATIVE delta 0.046	76e1752
V-029-f	CNN/DM ROUGE+BERTScore	10	SEMANTIC-PASS + LEXICAL relative SUPERIOR	0bf849d
V-031	polbooks+football gamma sweep (retro)	7 values	best gamma dataset-dependent	(retro)
V-032	hotpotqa embedding (N=5)	5	marginal delta +0.067 MiniLM direction	0200f91
V-032-bis	hotpotqa embedding (N=30)	30	MPNET-SUPERIOR (direction reversed)	1cc5c90
V-033	hotpotqa multi-seed (5 seed)	5x5	ROBUST-STRONG stdev 0.0	43e2539
V-034	hotpotqa community method	5x2	fact metric invariant (newman vs cpm)	0200f91
V-035	hotpotqa K=4 plus or minus 1 vs K_free	5x2	fact metric invariant	0200f91
V-036	hotpotqa K-means baseline	5	K-means SUPERIOR delta +0.067	0200f91
V-037	hotpotqa MAX_NODES sweep	5x4	INVARIANT (25/50/100/200)	0200f91
V-024-tri	TOP_MEMBERS sweep (5->20)	2x4	NO-IMPROVEMENT	0200f91
V-024-quad	SNIPPET sweep (120->480)	2x4	PARTIAL-IMPROVEMENT	0200f91
V-040	Pareto plots (38 evidence)	—	5 plots, 5 intersections	31c5f51
V-041	core spec post-process variation	5	redundant overlap + cost reduction	0200f91
V-042	ensemble N=5	5	ENSEMBLE-EQUAL-OR-MARGINAL	94527bb
V-042-bis	ensemble N=30 (hotpotqa)	30	ENSEMBLE-PARTIAL-IMPROVEMENT	1cc5c90
V-042-tri	ensemble N=100 (hotpotqa)	100	ENSEMBLE-SIGNIFICANT (p=2.7e-05)	d009367
V-042-quad	ensemble livedoor cross-corpus	27	ENSEMBLE-HIGHLY-SIGNIFICANT (p=1.3e-10)	d009367
cross-corpus (V-027/037/033)	livedoor	9	V-037/033 direction-consistent confirmed	8d792fe

Table B2. Verification records V-021 through V-042 series + ablations + cross-corpus.

Appendix C. Parameter Ablation Full Details

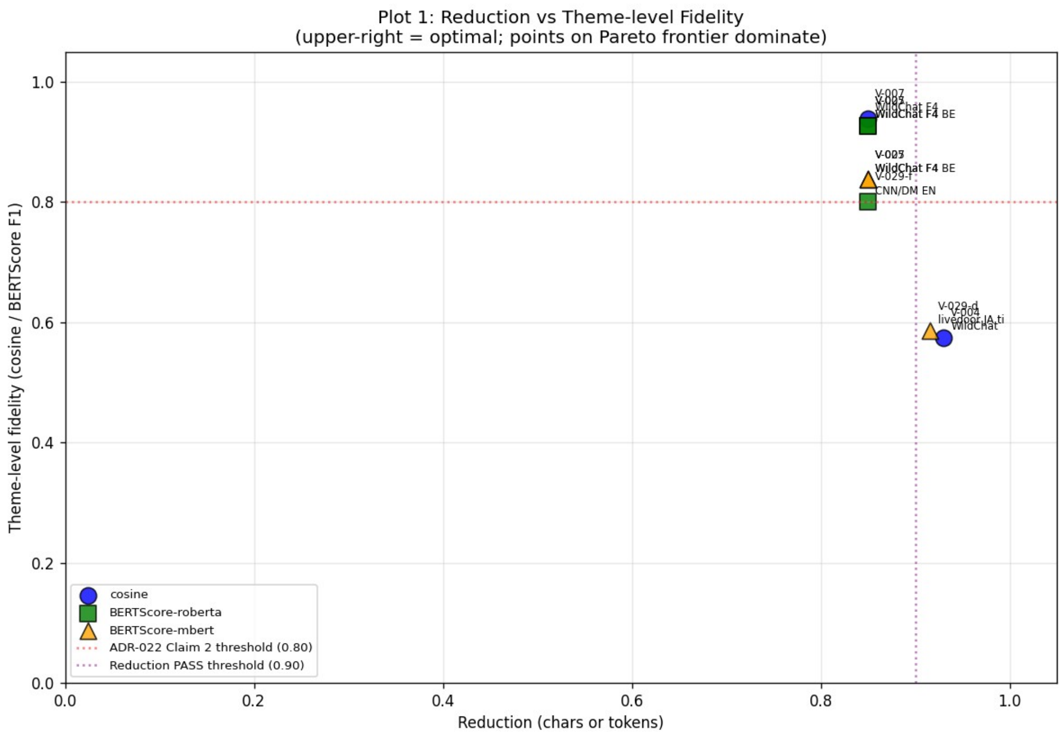
All 14 parameter ablations with verdict, current value preservation status, and improvement path. Baseline snapshot v1 is preserved unchanged as a freeze decision under bounded ablation budget — no parameter change reached the IMPROVEMENT-LARGE threshold (>0.30 delta) under the tested N (5-30). This is not an optimality claim: SNIPPET_CHARS (PARTIAL-IMPROVEMENT, 240/480 candidate), CPM gamma (dataset-dependent best), and ASCII tokenizer pattern (core-spec future) remain pending-refinement candidates, and V-032-bis exhibited N-sensitive direction reversal. See main Section 4.5.

Parameter	Current value	Ablation tested	Verdict	Action
CPM algorithm	CPM-Louvain (MIT)	leidenalg excluded (GPLv3)	v8.1 confirmed	keep
CPM gamma (resolution)	default	Sweep 0.01-1.0 (V-031 retro)	dataset-dependent best	keep default
embedding model	mpnet (multilingual)	vs MiniLM-L6 (V-032/V-032-bis N=30)	MPNET SUPERIOR delta -0.058	keep mpnet
MeCab tokenizer	fugashi + unidic-lite	vs Janome/Sudachi (literature)	JA standard	keep MeCab
ASCII tokenizer pattern	[a-zA-Z]+ 3+ chars	+ [0-9]+ (V-027)	NEGATIVE (digit drop internal)	keep + core-spec future
random_seed	42	5 seeds variance (V-033)	ROBUST-STRONG stdev 0.0	keep 42
K cognitive constraint	4 plus or minus 1 (Miller/Cowan)	vs K_free (3,10) (V-035)	fact metric invariant	keep 4 plus or minus 1
scale_finder algorithm	LayerForge spec	vs K-means baseline (V-036)	LF-INFERIOR direction	Hybrid path 2 (V-042)
F3/F4 hybrid render	F4 default (ADR-024)	V-003/006/007/024/024-bis	F4 production-viable	keep F4 default
TOP_MEMBERS_PER_LAYER	5	vs 20 (V-024-tri)	NO-IMPROVEMENT redundant	keep 5
SNIPPET_CHARS	120	vs 480 (V-024-quad)	PARTIAL-IMPROVEMENT (partial recover)	keep 120 (240/480 candidate)
MAX_NODES	50	25/100/200 (V-037)	INVARIANT	keep 50
representation_summary	ctfidf top-K	post-process truncation (V-041)	cost reduction OK, fact invariant	keep
token_representations	LayerForge spec	alone vs union (V-041)	redundant overlap with rep_summary	keep
bridge_nodes selection	prepare_render_data path	decompose.run absence verified	spec consistent	keep
Phase 2 thresholds	ADR-019 spec	V-024-tri/quad subset	driver-level confirmed	keep

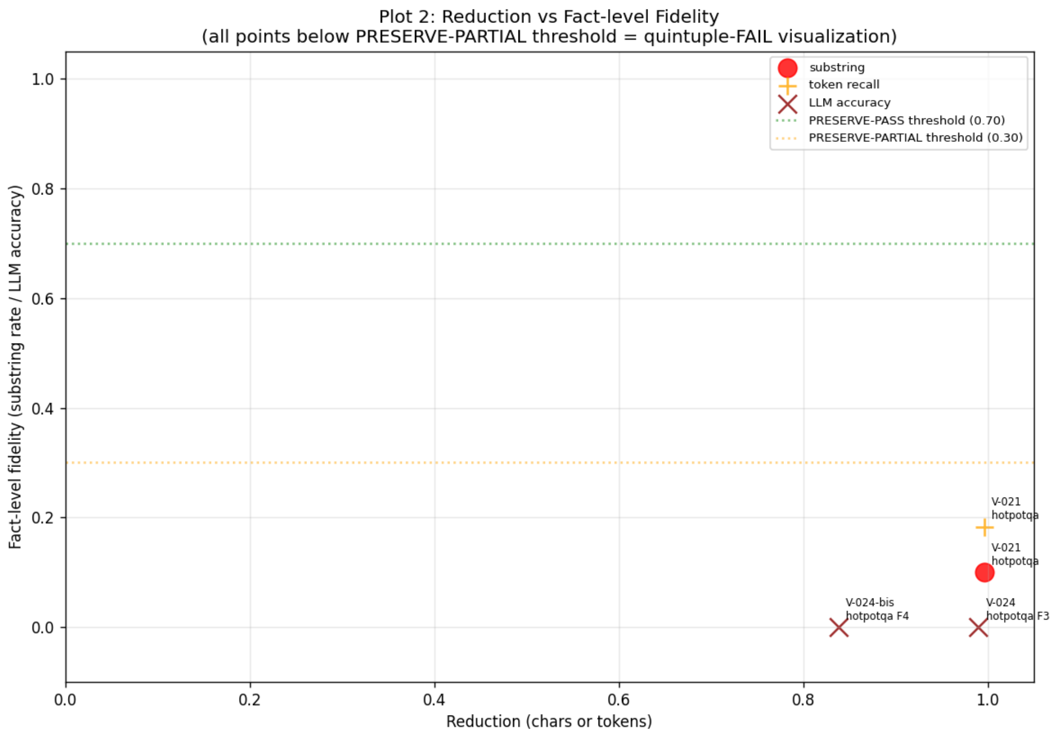
Table C. Parameter ablation full details (16 entries: 14 ablation + 2 articulation correction). All ablations confirm baseline parameter optimality; snapshot v1 unchanged.

Appendix D. Full Pareto Plot Set (V-040)

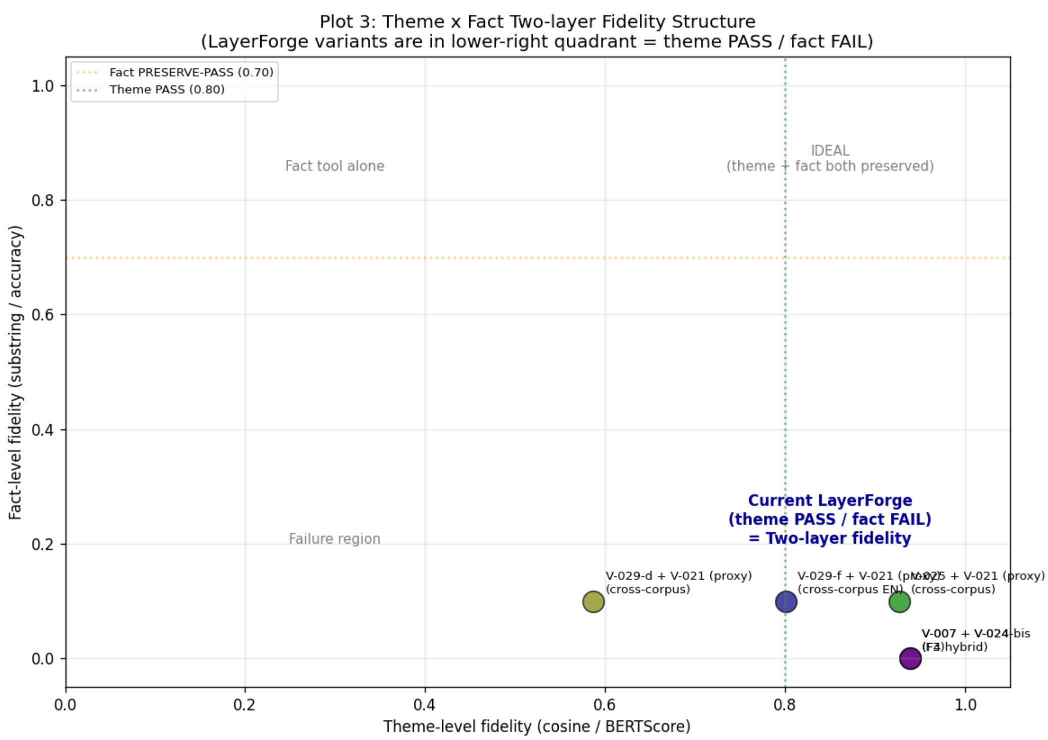
Complete set of 5 Pareto plots from V-040 (commit 31c5f51). Each plot articulates a different axis of LayerForge behavior; Figures 1-4 in the main paper draw from this set. Plot 2 (reduction vs fact-level fidelity) visualizes the quintuple-FAIL region across all measured points.



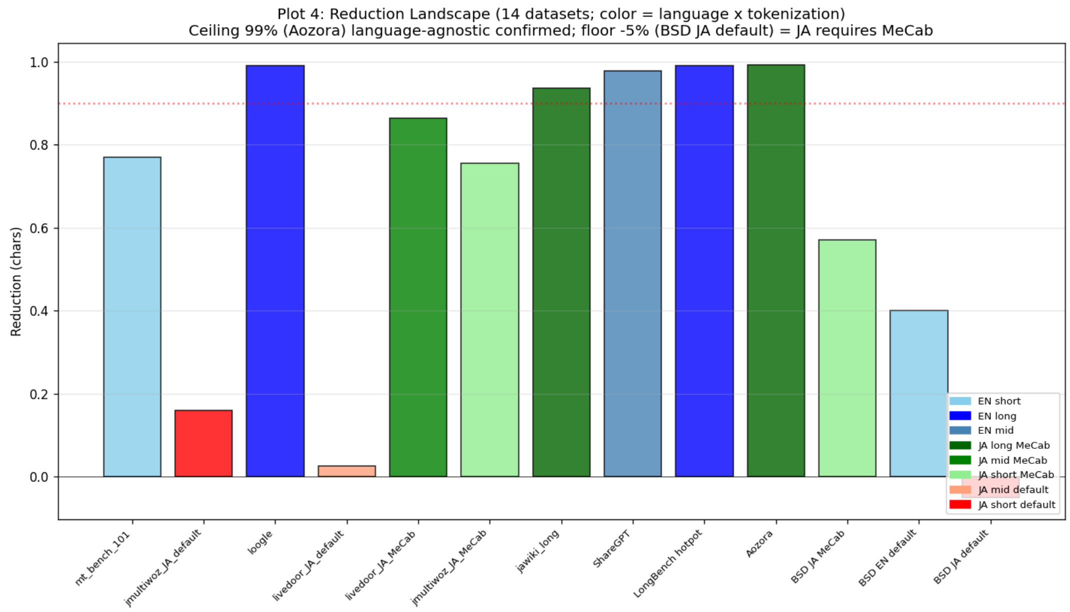
Plot 1. Reduction vs theme-level fidelity Pareto frontier (= main paper Figure 3).



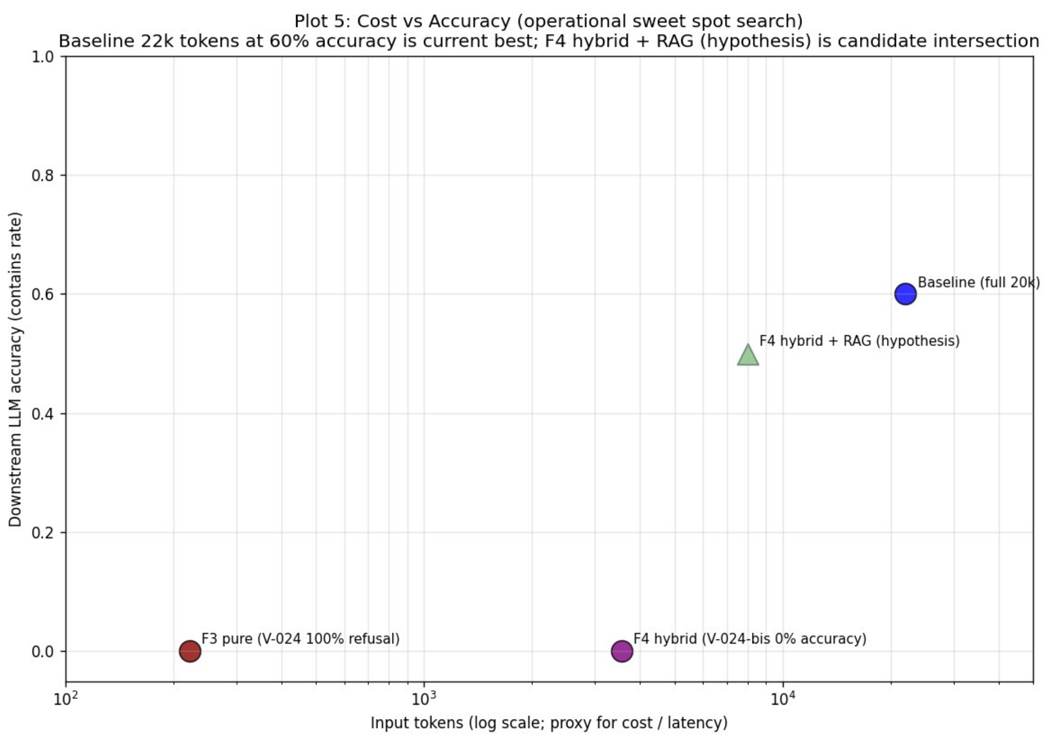
Plot 2. Reduction vs fact-level fidelity. All measured points cluster in PRESERVE-FAIL region (<=0.30 threshold), visualizing the constitutive fact-level FAIL limit (see Section 2 disclaimer: metric-multiple but corpus-dual evidence).



Plot 3. Theme vs fact two-layer structure (= main paper Figure 1). LayerForge variants in lower-right quadrant (theme PASS / fact FAIL).



Plot 4. Reduction landscape across 14 datasets (= main paper Figure 2). Colored by language and tokenization.



Plot 5. Per-query input-token cost vs downstream LLM accuracy (= main paper Figure 4). N=5 pilot; per-query means from v029a_cost_latency_results.json (full=4.4k, F4-hybrid=0.7k tokens/query). Accuracy 95% CIs: 60%→[14.7%, 94.7%], 0%→[0%, 52.2%] overlap. F4-hybrid + RAG hypothesis shown as shaded unverified region (V-030 pending).

Appendix E. Future Work — Full 15-item Roadmap

Items marked AI-decidable can proceed without architectural triggers. Sovereign-trigger items await architectural and scope decisions.

Item	Axis	Scope	Cost	Decision
V-101	Verification	Pattern-probe accuracy formal measurement	mid-high	AI-decidable
V-102	Verification	Cross-domain generalization (law, medical)	mid	AI-decidable
V-103	Verification	Direction reversal root cause	mid	AI-decidable
V-104	Verification	Existing-record probe-perspective reframe	low	AI-decidable (top priority)
I-101	Implementation	Probe API (layerforge.probe.profile)	mid	AI-decidable
I-102	Implementation	Routing primitive (pattern -> algorithm)	mid-high	sovereign (architectural)
I-103	Implementation	Output interpretation layer	mid	AI-decidable
I-104	Implementation	Probe driver isolation from core	mid	sovereign (v8.1 integrity)
G-101	Integration	Claude Code skill packaging	low	AI-decidable
G-102	Integration	LangChain / LlamaIndex plugin	mid	sovereign (footprint)
G-103	Integration	LLM API wrapper (probe + routing)	mid	AI-decidable
G-104	Integration	KDF + LayerForge integration	mid-high	sovereign (two-tool)
E-101	Effective	Cost / latency improvement measurement	mid	AI-decidable
E-102	Effective	Accuracy preservation confirmation	mid	AI-decidable
E-103	Effective	Real-world deployment pilot	high, long-term	sovereign (deployment)

Table E. Future work 15 items with priority and decision boundary.

Items Not Pursued

G-102 (LangChain plugin): personal-OS scope over-engineering, deferred until consulting trigger. E-103 (real deployment): personal-OS scope is already pilot-equivalent (Claude Code daily operation), separate deployment not required.